

Shubhayan Bagchi

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EDUCATION

- **Swami Vivekananda Institute of Science & Technology, Kolkata | MAKAUT** 2022 - 2026 (Expected)
Bachelor of Technology in Computer Science & Engineering
Coursework: Data Structure and Algorithm, Design & Analysis of Algorithm, DBMS, Operating Systems, Computer Networks, Compiler Design, Fullstack web development

TECHNICAL SKILLS

Languages: SQL, JavaScript, Typescript, Python, HTML, CSS, C

Frontend: React.js, Tailwind CSS, Shadcn, MUI, Bootstrap, Zustand, TanStack Query, Responsive Design

Backend: Node.js, Express.js, Flask, REST APIs, JWT & Cookie-based Auth, RBAC, Zod

Databases: PostgreSQL, MySQL, MongoDB, Database Design & Optimization

ORMs: Mongoose, Prisma

Tools & Technologies: Docker, Git, GitHub, Linux, Postman

PROJECTS

VagDevi.ai

College Tech Fest Winning Project

[Github](#)

- Developed a chatbot with text-to-text and image-to-text generation using the Gemini API.
- Integrated multilingual response translation and secure JWT-based authentication.
- Built with a modular React.js frontend, Flask backend, and MongoDB for data storage.

Tech Stack: React.js, Flask, Gemini API, MongoDB

College ERP System

Final Year Project | Team of 5

- Built a comprehensive College ERP System as a part of a 5-member team to centralize academic and administrative workflows, covering authentication and RBAC, admissions, academics, faculty management, attendance, examination & results, routines/timetables, fee management, and library management.
- Built all backend APIs for the Library Management module, including book catalog, issue/return workflow, overdue tracking, and library reports.
- Built all backend APIs for the Examination & Result Management module, including exam scheduling, marks entry, grade calculation, and transcription generation.
- Contributed backend support for all users dashboard reporting and analytics endpoints.
- Implemented JWT-based authentication and RBAC across student, faculty and admin roles.
- Deployed and demoed on a VPS during development; the college is currently evaluating the system for production adoption.

Tech Stack: PostgreSQL, Express.js, React.js, Node.js, Redis, Prisma, Docker, Nginx, Typescript, Tailwind CSS, Zustand, Tanstack Query, Zod, JWT

Swarlipi

Smart India Hackathon 2023 National Finalist

[Github](#)

- Built an AI-driven video dubbing pipeline as part of a team selected for the SIH 2023 national finale, transcribing speech with OpenAI Whisper, translating it via Google Translate, and generating dubbed audio with gTTs.
- Used pydub to time-stretch each generated audio segment so dubbed speech stays in sync with the original video's timing.
- Used moviepy to split the source video into per-segment clips, swap in translated audio, and reassemble the final dubbed video..

Tech Stack: Python, Flask, OpenAI Whisper, Google Translate API, gTTs, moviepy, pydub

ACHIEVEMENTS

- **SIH'23 finalist:** Participated in Smart India Hackathon 2023 finale at CGC Landran, Punjab Dec 19-20, 2023

POSITION OF RESPONSIBILITY

- **Group Leader (Coding Club, Swami Vivekananda Institute of Science & Technology, Kolkata)**
Responsibilities: Mentoring juniors in the field of fullstack web development